LABORATORY OBSERVATION / RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

EXPERIMENT NO.: 3

JavaScript Arrays and Functions

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1. TITLE

Write a program using JavaScript (arrays, functions)

2. AIM

To write and execute a JavaScript program that demonstrates arrays, array operations, regular functions and arrow functions.

3. OBJECTIVE

- To create and initialize an array in JavaScript.
- To find the length of an array and access individual elements.
- To modify an array element.
- To understand push(), pop(), unshift() and shift() methods.
- To define and call functions using arrays as parameters.
- To find the highest and lowest marks using functions.
- To understand arrow functions with zero, one and multiple parameters.

5. THEORY

An array in JavaScript is used to store multiple values in a single variable. Array elements are accessed using an index starting from 0. JavaScript provides built-in properties and methods for working with arrays.

- length – returns the number of elements.
- Indexing – accesses individual elements.
- push() – adds an element at the end.
- pop() – removes the last element.
- unshift() – adds an element at the beginning.
- shift() – removes the first element.

A function is a reusable block of code that performs a particular task. This program uses regular functions to display marks and find the highest and lowest values. It also demonstrates arrow functions with different parameter forms.

6. ALGORITHM / PROCEDURE

1. Create an array named marks with five values.

2. Display the original array and its length.

3. Access the first, third and last elements.

4. Change the third element.

5. Use push(), pop(), unshift() and shift() to modify the array.

6. Define showMarks() to display all elements.

7. Define findHighest() and findLowest() to find extreme values.

8. Define and call arrow functions for welcome, doubling and multiplication.

9. Call the functions and display the results in the console.

HOW TO RUN THE PROGRAM

1. Open the JavaScript file in Visual Studio Code.

2. Make sure Node.js is installed on the system.

3. Open the VS Code Terminal using Terminal → New Terminal.

4. Run the program using the command: node filename.js

5. The output will be displayed in the VS Code terminal.

Paste VS Code Terminal screenshot showing the execution command and output here

```
PS F:\full-stack-web-development> cd F:\full-stack-web-development\02-Record\week-03-jsarrayFunctions
PS F:\full-stack-web-development\02-Record\week-03-jsarrayFunctions> node array.js
```

7. PROGRAM

Complete JavaScript Source Code

```javascript
// Create an array
let marks = [75, 82, 68, 90, 85];
console.log("Original Array:", marks);
// 1. Length
console.log("Number of Elements:", marks.length);
// 2. Accessing Array Elements
console.log("First Mark:", marks[0]);
console.log("Third Mark:", marks[2]);
console.log("Last Mark:", marks[marks.length - 1]);
// 3. Changing an Element
marks[2] = 72;
console.log("After Changing Third Element:", marks);
// 4. push() - Add element at the end
marks.push(95);
console.log("After push():", marks);
// 5. pop() - Remove last element
marks.pop();
console.log("After pop():", marks);
// 6. unshift() - Add element at the beginning
marks.unshift(70);
console.log("After unshift():", marks);
// 7. shift() - Remove first element
marks.shift();
console.log("After shift():", marks);

// Function to display all elements
function showMarks(arr) {
    console.log("Student Marks:");
    for (let i = 0; i < arr.length; i++) {
        console.log("Mark", i + 1, ":", arr[i]);
    }
}
// Function to find the highest mark
```

```
function findHighest(arr) {
    let highest = arr[0];
    for (let i = 1; i < arr.length; i++) {
        if (arr[i] > highest) {
            highest = arr[i];
        }
    }
    return highest;
}
// Function to find the lowest mark
function findLowest(arr) {
    let lowest = arr[0];
    for (let i = 1; i < arr.length; i++) {
        if (arr[i] < lowest) {
            lowest = arr[i];
        }
    }
    return lowest;
}
// Arrow function without parameters
const welcome = () => {
    console.log("Welcome to JavaScript Arrays");
};
welcome();
// Arrow function with one parameter
const double = value => value * 2;
console.log("Double of 15:", double(15));
// Arrow function with multiple parameters
const multiply = (x, y) => {
    return x * y;
};
console.log("Multiplication of 6 and 7:", multiply(6, 7));

// Calling functions
showMarks(marks);
console.log("Highest Mark:", findHighest(marks));
console.log("Lowest Mark:", findLowest(marks));
```

8. CODE EXPLANATION

Code / Concept	Purpose
marks.length	Returns the number of array elements.
marks[index]	Accesses an element using its index.
push()	Adds an element at the end.
pop()	Removes the last element.
unshift()	Adds an element at the beginning.
shift()	Removes the first element.
showMarks(arr)	Displays all marks using a loop.
findHighest(arr)	Returns the highest mark.
findLowest(arr)	Returns the lowest mark.
Arrow functions	Provide shorter syntax for defining functions.
console.log()	Displays output in the JavaScript console.

9. TEST CASES AND EXECUTION DATA

Fill the actual result/status after executing the program.

Test Case	Test / Input	Expected Result	Actual Result / Status
TC-01	Create array	Original array should be displayed correctly	To be filled
TC-02	Check length/indexing	Length and requested elements should be correct	To be filled
TC-03	Modify array	push(), pop(), unshift() and shift() should update the array correctly	To be filled
TC-04	showMarks()	All marks should be displayed	To be filled
TC-05	findHighest()	Highest mark should be 90	To be filled
TC-06	findLowest()	Lowest mark should be 72 after the changes	To be filled
TC-07	Arrow functions	Welcome message, 30 and 42 should be displayed	To be filled

10. OUTPUT

Paste the actual JavaScript console screenshot below.

Output 1: Array Operations

```
Original Array: [ 75, 82, 68, 90, 85 ]
Number of Elements: 5
First Mark: 75
Third Mark: 68
Last Mark: 85
After Changing Third Element: [ 75, 82, 72, 90, 85 ]
After push(): [ 75, 82, 72, 90, 85, 95 ]
After pop(): [ 75, 82, 72, 90, 85 ]
After unshift(): [ 70, 75, 82, 72, 90, 85 ]
After shift(): [ 75, 82, 72, 90, 85 ]
```

This is the output demonstrating arrays.

Output 2: Functions and Arrow Functions

```
This line demonstarates the function without parameters.
Double of 15: 30
Multiplication of 6 and 7: 42
Student Marks:
Mark 1 : 75
Mark 2 : 82
Mark 3 : 72
Mark 4 : 90
Mark 5 : 85
Highest Mark: 90
Lowest Mark: 72
```

This is the output demonstrating functions.

11. RESULT

Thus, a JavaScript program was written to demonstrate arrays, array methods, regular functions and arrow functions. The program performs array manipulation and calculates the highest and lowest marks.

12. CONCLUSION

The experiment provides a basic understanding of JavaScript arrays, array operations, functions and arrow functions.